

Varun Keshav Kumar

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Professional Summary

Software Engineer with 4+ years building data pipelines and automation for enterprise clients, transitioning into Data Science. Designed and evaluated a zero-shot computer vision annotation pipeline (Grounding DINO + SAM2) on the Indian Driving Dataset, reaching 0.55 mIoU and 0.40 AP@0.5 against strongly supervised baselines – published as a co-authored conference paper. Built Python audit logic to detect fee discrepancies across 72 client accounts monthly, and Snowflake pipelines powering operational business dashboards. Pursuing an M.Tech in Data Science and Artificial Intelligence at PES University.

Technical Skills

Data Science & ML: Python, SQL, R, pandas, NumPy, Scikit-learn, Keras, OpenCV, Statistical Modelling, Semantic Segmentation

GenAI & LLMOps: LLMs, Retrieval-Augmented Generation (RAG), Vector Databases (FAISS, Chroma, Pinecone), Grounding DINO, SAM/SAM2, Zero-Shot Inference, Prompt Engineering

Data Engineering: Snowflake, AWS (EMR, S3, Lambda, SQS, EKS, ECS), Apache Kafka, SQL query optimisation, ETL pipelines

Databases: Snowflake, MySQL, PostgreSQL, Vector Databases

Engineering & Automation Tools: Java, Scala, JavaScript, TypeScript, Flask, Spring Boot, React.js, Selenium, Retool, Camunda, Docker, Kubernetes, Git, CI/CD, Linux, Agile/Scrum

Projects and Publications

- **Automated Data Annotation for the Indian Driving Dataset** – Co-authored conference paper (PES University, DSAI). Designed a two-stage zero-shot annotation pipeline combining Grounding DINO (language-conditioned object detection) and SAM/SAM2 (instance segmentation) across the IDD's 46,588 detection images and 10,003 segmentation images (34 object classes). Benchmarked against strongly supervised baselines on the validation split: achieved 0.5527 mIoU (vs. 0.5541 for a supervised ERFNet baseline) and 0.4029 AP@0.5 (vs. 0.4991 for a supervised ResNet101 baseline), demonstrating that zero-shot foundation models can approach task-specific supervised performance on complex, unstructured Indian road scenes without any dataset-specific training.
- **Mask Detection System** – Real-time computer vision classifier using MobileNetV2 and OpenCV. Code: github.com/kvarun314.

Professional Experience

Analyst – Software Engineer

Jul 2026 – Present

Endava Solutions India Private Ltd., Bengaluru, India

Transferred from Omnicom Media Group – Flywheel/Omni following Omni's strategic partnership with Endava, which moved Flywheel/Omni Product & Technology into Endava's operating model; same role and responsibilities.

Analyst – Software Engineer

Apr 2026 – Jun 2026

Omnicom Media Group – Flywheel, Bengaluru, India

- Built a distributed Redis + MySQL RDS caching layer with SHA-256 content fingerprinting, cutting duplicate WFL case submissions by 88% across 10,400+ ASIN PDP change cases.
- Cut database round-trips from 1,000 queries to a single call via Redis batch reads and pipelined writes; added async, non-blocking cache writes to keep case creation unaffected.
- Contributed to Flywheel's infrastructure migration to Omni: data pipeline refactoring, backend authentication changes, and architectural migration to PostgreSQL and Snowflake.

Senior Associate Software Developer**Apr 2025 – Mar 2026***Omnicom Media Group – Flywheel, Bengaluru, India*

- Designed and deployed end-to-end automation for 8 fee recovery reports (incl. cooperative audit calculations) on AWS EKS, eliminating manual processing.
- Integrated backend APIs with dashboard microservices and React components to surface fee-recovery insights across 72 client accounts monthly, informing roughly 10 new recovery opportunities per month projected at \$10K–\$100K+ per client depending on invoice volume.
- Built S3-backed data download workflows via Retool APIs, improving analyst self-service and cutting data access latency.
- Automated data generation and verification pipelines on AWS ECS, improving deployment reliability and shortening release cycles.

Associate Software Developer**Jul 2024 – Apr 2025***Omnicom Media Group – Flywheel, Bengaluru, India*

- Authored Python audit logic to detect fee discrepancies in Amazon-seller client invoices, standardising fee recovery calculations across accounts and eliminating manual verification; built Selenium automation to source 1,000+ invoice downloads per run across clients and regions.
- Built Snowflake pipelines feeding operational dashboards with refreshed metrics and configurable alerts for the operations team.
- Developed chargeback report APIs with Snowflake integration, delivering accurate, timely financial data to stakeholders.

Software Development Engineer**Feb 2022 – Jul 2024***Jio Platforms Ltd., Bengaluru, India*

- Contributed to the Network Stack Assurance Layer (HP Service Manager integration); designed and delivered 10 Java microservice applications to Solutions Architect specifications, and remediated security vulnerabilities across 18 Java applications for enterprise compliance.
- Deployed a full-stack open-source ITSM tool for the IAX instance and scaled it to a second Reliance Jio Network Stack instance.

Education**M.Tech, Data Science and Artificial Intelligence****Expected 2026***PES University, Bengaluru, India***B.Tech, Information Science and Engineering****2014 – 2018***School of Engineering and Technology, Jain University, Bengaluru, India*

GPA: 8.22 / 10